



PHASE 5

AI Integration and Budget Safety

Prompt controls, deterministic normalization and failure handling

SMARTPOCKET AI • SMART BUDGET AND RECOMMENDATION ASSISTANT

1. AI Integration and Budget Safety

1.1 Prompt Contract

- Assign a clear planner role and Indian market context.
- Include sanitized user inputs as structured data.
- Specify required JSON keys and planner-specific fields.
- Require numeric INR values and category-level totals.
- Limit icons and output fields to known values.
- Require search URLs rather than unsupported claims about specific product availability.

1.2 Deterministic Normalization

The backend converts a candidate AI response into a trusted application response. It maps alternate keys, replaces unsupported values, ensures non-negative prices and positive quantities, recalculates each category, and recomputes the overall spent and remaining values.

Budget scaling logic

```
def fit_to_budget(items, budget):
    spent = sum(max(0, item["estimated_price"]) for item in items)
    if spent <= budget or spent == 0:
        return items
    scale = budget / spent
    for item in items:
        item["estimated_price"] = round(item["estimated_price"] * scale)
    # Remove any rounding excess from the final priced item.
    excess = sum(i["estimated_price"] for i in items) - budget
    if excess > 0:
        items[-1]["estimated_price"] -= excess
    return items
```

1.3 Error Handling

Condition	Status	Client behavior
Missing AI configuration	503	Show service configuration message
AI timeout	502	Show retryable timeout message
Invalid AI JSON	502	Reject raw output and show a clean failure
No budget categories	502	Do not render an incomplete plan
Invalid uploaded image	422	Ask for a valid supported image



Condition	Status	Client behavior
Invalid form values	422	Highlight validation issue
Missing or expired token	401	Return to authentication flow

Guiding principle: generative recommendations remain advisory, and every value used by the product is validated by deterministic application logic.